

Center transport under bounded point-reflection symmetry

MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection

theorem

Uses a doubling argument to identify the reflection centers under a bijective isometry.

Statement

Let X, Y be real normed affine spaces, with translation spaces V, W . For $c \in X$, point reflection about c is $\sigma_c(x) = c - (x - c)$; define σ_d on Y similarly. Let $S \subseteq X$, $T \subseteq Y$, choose $c \in S$ and $d \in T$, and let $f : S \rightarrow T$ be a bijective isometry. If S is bounded and $\sigma_c(S) \subseteq S$, $\sigma_d(T) \subseteq T$, then $f(c) = d$.

Assumptions

The metrics come from the normed translation spaces V, W . The centers belong to their respective subsets; S is bounded; and each subset is invariant under reflection about its designated center. No convexity, openness, completeness or finite-dimensionality assumption is used.

Conclusion

The given isometry sends the designated source center to the designated target center: $f(c) = d$.

Proof route

A conjugation argument doubles the displacement of a center under a self-isometry. Boundedness forces that displacement to vanish; applying the result to the transported target reflection identifies the two centers.

▼ Proof steps

1. Point reflection is an involutive isometry and fixes its center. Hence the two invariance assumptions restrict σ_c and σ_d to bijective isometries of S and T ; this is Restriction of point reflection.
2. For a bijective self-isometry $g : S \rightarrow S$, define

$$\Gamma(g) = \sigma_c \circ g^{-1} \circ \sigma_c \circ g.$$

Since $\sigma_c(c) = c$, distance preservation gives

$$\begin{aligned} \text{dist}(\Gamma(g)(c), c) &= \text{dist}(g^{-1}(\sigma_c(g(c))), c) \\ &= \text{dist}(\sigma_c(g(c)), g(c)) \\ &= 2 \text{dist}(g(c), c). \end{aligned}$$

This is the doubling calculation in Bounded reflection-center lemma.

3. The set of numbers $\text{dist}(g(c), c)$ is bounded because both points lie in bounded S . Let C be its supremum. Since $\Gamma(g)$ is again a self-isometry of S , Step 2 yields $2 \text{dist}(g(c), c) \leq C$ for every g , hence $2C \leq C$ and therefore $C = 0$. Thus every self-isometry of S fixes c .
4. Apply the preceding result to $g = f^{-1} \circ \sigma_d \circ f$. Then

$$f^{-1}(\sigma_d(f(c))) = c,$$

so $\sigma_d(f(c)) = f(c)$. A point is fixed by reflection about d only when it equals d ; equivalently, $f(c) - d = -(f(c) - d)$. Hence $f(c) = d$.

Main citations

- Exact theorem and proof
- Bounded reflection-center lemma
- Restriction of point reflection

Lean source signature (exact)

```
theorem map_center_of_mapsTo_pointReflection
  {s : Set P} {t : Set Q} {c : P} {d : Q} (f : s ≈i t)
  (hc : c ∈ s) (hd : d ∈ t) (hs : IsBounded s)
  (hsreflect : MapsTo (pointReflection ℝ c) s s)
  (htreflect : MapsTo (pointReflection ℝ d) t t) :
  f ⟨c, hc⟩ = ⟨d, hd⟩
```

In the source	Mathematical meaning
$\{s : \text{Set } P\} \{t : \text{Set } Q\}$	The source subset $S \subseteq X$ and target subset $T \subseteq Y$. The Lean names are s, t and the ambient affine spaces are P, Q .
$\{c : P\} \{d : Q\}$	The designated centers $c \in X$ and $d \in Y$.
$(f : s \approx_i t)$	The distance-preserving bijection $f : S \rightarrow T$, including its inverse.
$(hc : c \in s) (hd : d \in t)$	The two membership assumptions which allow c and d to be used as points of the subsets.
$(hs : \text{IsBounded } s)$	The source set S is bounded.
$\text{MapsTo } (\text{pointReflection } \mathbb{R} \ c) \ s \ s$	For every $p \in S$, the point $r_c(p) = c - (p - c)$ belongs to S .
$\text{MapsTo } (\text{pointReflection } \mathbb{R} \ d) \ t \ t$	For every $q \in T$, $r_d(q) \in T$.

In the source	Mathematical meaning
$f \langle c, hc \rangle = \langle d, hd \rangle$	The conclusion $f(c) = d$ in T . The brackets attach membership proofs to the same points; they do not choose new centers.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

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